ICPC + GSoC + Google SWE Roadmap

Sem 1 se Sem 8 tak Complete Plan

Contents

Roadmap: ICPC + GSoC + Google SWE

CSE Student — Sem 1 se Sem 8 tak ka Complete Plan

Tumhare Goals

1. ICPC me strong performance (Sem 3-4 tak peak level ready)
2. GSoC selection
3. Sem 4 ke baad Internship
4. Sem 7-8 tak Google SWE job

Folder Structure

File	Kya hai isme
`01_Year1_Foundation.md`	Sem 1-2: C++ + Basic DSA
`02_Year2_ICPC_GSoC.md`	Sem 3-4: ICPC peak + GSoC applications
`03_Year3_Internship_SWE.md`	Sem 5-6: Internship + SWE prep start
`04_Year4_Google_Prep.md`	Sem 7-8: Google interview prep + applications
`05_Resources.md`	Saare important links/platforms ek jagah
`06_Weekly_Checklist.md`	Har hafte follow karne wala tracker

Golden Rules (Poore 4 saal ke liye)

1. **Consistency > Intensity** — Roz 1-2 hours CP better hai weekend pe 8 hours se

2. **Track everything** — Codeforces rating, LeetCode count, GitHub contributions — sab likho
3. **Team practice ICPC ke liye non-negotiable hai** — solo practice se team synergy nahi banti
4. **GSoC saal me sirf ek chance** — March deadline miss mat karo, December se hi prep shuru karo
5. **Burnout se bachna hai** — har mahine 2-3 din full rest lo

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Reality Check (Important!)

- ICPC World Finals bohot competitive hai — regional se qualify karna hi bada achievement hai, ye realistic milestone samjho
- GSoC selection rate low hai — quality proposal + pehle se contribution karna zaroori hai
- Google job — ICPC/GSoC achievements resume ko strong banayenge, lekin interview (LeetCode + System Design + CS fundamentals) alag se prepare karna padega

Is roadmap ko rigid rule mat samjho — apni situation ke hisaab se adjust karte raho. # Year 1 (Sem 1-2): Foundation — C++ + Basic DSA

Month 1-2: C++ Crash Course

Python pata hai to fast jaoge, bas syntax difference samajhna hai.

- ☐ Basic syntax, data types, loops, functions
- ☐ Pointers & references (ye Python me nahi hota, dhyan se seekho)
- ☐ OOP basics (classes, inheritance)
- ☐ STL Mastery:
    - vector, pair, string
    - map, unordered_map, set, unordered_set
    - priority_queue, queue, stack
    - sort(), custom comparators

Resource: Love Babbar C++ playlist / CodeWithHarry (Hindi)

Month 3-6: Core DSA Topics

Order follow karo, skip mat karo:

1. Arrays & Strings (2 weeks)
2. Recursion & Backtracking (2 weeks)
3. Basic Trees + BST (2 weeks)
4. Basic Graphs — BFS/DFS (2 weeks)
5. Sorting & Searching algorithms (1 week)
6. Basic Dynamic Programming (2 weeks)

Practice routine: - Codeforces Div 3/Div 4 — 1 contest/week - Daily 1-2 problems (rating 800-1200)

Target: Codeforces rating 1200-1400 by end of Sem 1

Month 7-12 (Sem 2): Level Up

- ☐ Dynamic Programming (intermediate) — knapsack, LIS, digit DP intro
- ☐ Graphs advanced — Dijkstra, Union-Find (DSU), MST (Kruskal/Prim)
- ☐ Number theory — GCD/LCM, modular arithmetic, sieve of Eratosthenes
- ☐ Bit manipulation basics

CP Practice: - CSES Problem Set — start solving sequence-wise - Codeforces Div 2 contests — 1/week - Virtual contests for practice

ICPC prep starts: - [ ] Team bana lo (3 members, complementary strengths) - [ ] Weekly team practice session start karo - [ ] College CP club join/start karo

GSoC groundwork (abhi se shuru karo, apply baad me karoge): - [ ] Git & GitHub properly seekho (branch, commit, PR, merge conflicts, rebase) - [ ] Ek domain choose karo jisme interest ho: Web Dev / ML / Systems / DevOps - [ ] GitHub explore karna start karo — "good first issue" tag dekhna shuru karo

Target: Codeforces rating 1500-1600 by end of Sem 2

Sem 1-2 Checklist Summary

- ☐ C++ fully comfortable
- ☐ 150+ problems solved (Codeforces + CSES combined)
- ☐ Codeforces rating 1500+
- ☐ ICPC team formed
- ☐ Git/GitHub comfortable
- ☐ Open source domain chosen # Year 2 (Sem 3-4): ICPC Peak + GSoC — Crunch Period

Ye tumhara sabse busy phase hai — dono cheez saath chalengi. Time management critical hai.

ICPC — Advanced Prep

Sem 3 Topics:

- ☐ Advanced DP — bitmask DP, tree DP, DP on graphs
- ☐ Segment Trees (with lazy propagation)
- ☐ Fenwick Tree (Binary Indexed Tree)
- ☐ Advanced Graphs — SCC, bridges/articulation points, flow algorithms (basic)

☐ String algorithms — KMP, Z-function, hashing

Sem 4 Topics:

☐ Heavy-Light Decomposition
☐ Advanced flow algorithms (max flow, min cut)
☐ Suffix arrays basics
☐ Game theory basics
☐ Advanced number theory (CRT, Euler's totient)

Team Practice Routine:

- 2x weekly team contests (Codeforces Gym / past ICPC Regional sets)
- Post-contest: upsolve together, discuss approach differences
- Track weak areas as a team, assign focused practice

Milestone:

☐ Participate in ICPC Regional Qualifier
☐ Target Codeforces rating: **1800-2000+ (Expert/Candidate Master)**

Reminder: World Finals bahut competitive hai — regional level me strong hona hi is phase ka realistic target hai.

GSoC — Application Cycle

Timeline (roughly, check official gsoc.dev site every year): - Dec-Feb: Organizations research + community engagement - Feb-March: Org list announced officially - March: Applications open, proposals submit - May: Results announced

Step-by-step:

1. **Dec-Jan: Shortlist 2-3 organizations**
   - Apna chosen domain (web/ML/backend etc.) match karo
   - Past year projects dekho unke
2. **Jan-Feb: Community engagement**
   - Discord/Slack/mailing list join karo
   - Introduce yourself, ask how to start contributing
   - "Good first issue" solve karo — chhote bug fixes, docs improvement
   - PRs bhejo, mentors se feedback lo
3. **Feb-March: Proposal writing**
   - Technical approach clearly likho
   - Week-by-week timeline do (12+ week coding period)
   - Past contributions/PR links include karo
   - Feedback lo kisi senior/mentor se before submitting
4. **March: Submit**

- Multiple orgs me apply kar sakte ho, but sirf ek project accept hoga
- Quality > quantity — 2-3 achhe proposals better hain 10 generic se

Eligibility note: Ab students-only nahi hai — koi bhi beginner (2 saal se kam open source experience) eligible hai, so competition high hai.

Internship Prep Starts (End of Sem 4)

- ☐ LeetCode grind start karo — NeetCode 150 pattern-wise
- ☐ Resume banao — ICPC + GSoC (agar select hua) + projects
- ☐ LinkedIn profile active karo, connections banao
- ☐ Apply karna start karo summer internships ke liye (Sem 4 khatam hote hi)

Year 2 Checklist Summary

- ☐ ICPC Regional participate kiya
- ☐ Codeforces rating 1800+
- ☐ GSoC me apply kiya (kam se kam ek attempt)
- ☐ 2-3 merged PRs kisi open source project me
- ☐ LeetCode 100+ problems solved
- ☐ Resume ready # Year 3 (Sem 5-6): Internship + SWE Foundation

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Internship (Primary Goal Isi Year Ka)

Application Strategy:

- ☐ On-campus placement cell se apply karo
- ☐ Off-campus: LinkedIn, company career pages, referrals
- ☐ Cold outreach — GSoC/ICPC mentors, seniors se referral maango
- ☐ Apply broadly — startups + mid-size + big tech, sab try karo

Resume highlights (isi order me priority):

1. ICPC achievements (rating, regional participation)
2. GSoC (agar mila) ya open source contributions
3. Strong projects (2-3, well-documented on GitHub)
4. Relevant coursework

LeetCode / DSA Interview Prep

- ☐ NeetCode 150 complete karo (agar Sem 4 me nahi kiya to abhi)

- ☐ Pattern-wise practice: Two pointers, Sliding window, Binary search, DP, Graphs, Trees
- ☐ Medium level problems focus — 300+ target by end of Sem 6
- ☐ Mock interviews start karo (Pramp, interviewing.io — free platforms)

CS Fundamentals (Interviews me poochte hain)

- ☐ Operating Systems — process/threads, deadlock, memory management, scheduling
- ☐ DBMS — normalization, indexing, transactions, SQL queries
- ☐ Computer Networks — OSI model, TCP/UDP, HTTP/HTTPS basics
- ☐ OOP concepts — deep dive (polymorphism, encapsulation, design patterns basics)

Resource suggestion: Gate Smashers (Hindi) for quick fundamentals revision

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System Design — Basics Start Karo

- ☐ Scalability basics — horizontal vs vertical scaling
- ☐ Load balancers, caching concepts
- ☐ Database basics — SQL vs NoSQL, when to use what
- ☐ Watch: Gaurav Sen YouTube series (beginner-friendly)

Deep system design Sem 7-8 me karoge — abhi sirf foundation.

ICPC Continuation (agar abhi bhi eligible ho)

- Team practice continue rakho agar ICPC eligibility bachi hai
- Regional attempts continue karo

Year 3 Checklist Summary

- ☐ Summer internship secure kiya
- ☐ LeetCode 300+ problems
- ☐ CS fundamentals revised
- ☐ System design basics clear
- ☐ Internship se real-world experience gained # Year 4 (Sem 7-8): Google SWE — Final Push

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LeetCode — Advanced Phase

- ☐ Google-tagged problems specifically practice karo (LeetCode has company tags in premium, ya GFG/other lists dhundo)
- ☐ Medium-Hard ratio badhao — 60% Medium, 40% Hard
- ☐ Timed practice — 45 min me ek problem solve karne ki habit banao (interview simulation)
- ☐ Revision — pehle solve kiye hue problems dobara karo (spaced repetition)

System Design — Deep Dive

- ☐ Scalability patterns — sharding, replication, consistent hashing
- ☐ Caching strategies — Redis, CDN concepts
- ☐ Database design — SQL vs NoSQL tradeoffs, when to use which
- ☐ Real system design case studies — URL shortener, rate limiter, chat app, news feed
- ☐ Practice explaining design out loud (mock interviews)

Mock Interviews (Critical Phase)

- ☐ Pramp / interviewing.io — weekly mock interviews
- ☐ Seniors/GSoC-ICPC network se mock lo agar possible ho
- ☐ Behavioral questions prep karo (STAR method — Situation, Task, Action, Result)
- ☐ "Tell me about a time…" questions ke liye apne ICPC/GSoC/internship experiences se stories ready rakho

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Application Strategy

On-campus:

- ☐ Google agar campus drive karta hai college me — apply zaroor karo
- ☐ Placement cell se timeline follow karo

Off-campus / Referral:

- ☐ LinkedIn pe Google employees se connect karo (especially GSoC/ICPC alumni network)
- ☐ Referral maangna normal hai — politely reach out karo
- ☐ Google Careers page directly apply bhi karte raho

Resume final polish:

- ICPC (rating, achievements)

- GSoC (project, org, outcome)
- Internship experience (impact-focused, numbers ke saath)
- Strong projects

Interview Rounds (Typical Google Process)

1. **Online Assessment / Phone Screen** — 1-2 coding rounds
2. **Onsite/Virtual Onsite** — usually 4-5 rounds:
   - 2-3 coding rounds (DSA)
   - 1 system design round (for SWE roles, sometimes for new grads too)
   - 1 “Googleyness” / behavioral round
3. **Hiring Committee Review**
4. **Team Matching**

Year 4 Checklist Summary

- ☐ LeetCode 500+ problems total (cumulative)
- ☐ System design comfortable at mid-level
- ☐ 10+ mock interviews done
- ☐ Resume finalized with strong story
- ☐ Applied to Google (on-campus + off-campus both)
- ☐ Referral network built # Resources — Sab Ek Jagah

C++ Learning

- Love Babbar C++ Playlist (YouTube, Hindi)
- CodeWithHarry C++ Course (YouTube, Hindi)
- cplusplus.com / cppreference.com (documentation)

Competitive Programming

- **Codeforces** — codeforces.com (main practice platform, ratings track karo)
- **CSES Problem Set** — cses.fi/problemset (structured, sequence-wise problems)
- **CP-Algorithms** — cp-algorithms.com (advanced technique explanations)
- **AtCoder** — atcoder.jp (additional practice, good problem quality)
- **Codeforces Gym** — past ICPC regional problems yahan milte hain

Strivers A2Z DSA Sheet

- YouTube “take U forward” channel — structured DSA sheet, bahut popular hai beginners ke liye

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ICPC Specific

- icpc.global — official website, regionals ka schedule
- ICPC India regional info — apne region ka specific page dekho
- Past year ICPC problems — Codeforces Gym pe milte hain

GSoC

- summerofcode.withgoogle.com — official website
- Organizations list — har saal Feb-March me publish hoti hai
- GitHub "good first issue" — good-first-issue.github.io

Git & GitHub

- "Git and GitHub for Beginners" (YouTube — freeCodeCamp)
- GitHub Docs — docs.github.com

Internship/Job Prep

- **LeetCode** — leetcode.com
- **NeetCode 150** — neetcode.io (pattern-wise curated list)
- **Pramp** — pramp.com (free mock interviews)
- **interviewing.io** — mock interviews with engineers

CS Fundamentals

- Gate Smashers (YouTube, Hindi) — OS, DBMS, CN quick revision
- Neso Academy (YouTube) — detailed CS fundamentals

System Design

- Gaurav Sen (YouTube) — beginner-friendly system design
- "System Design Interview" by Alex Xu (book, jab ready ho to padhna)
- ByteByteGo (YouTube) — visual explanations

Communities (Join karo, help milegi)

- Codeforces Discord/Telegram groups
- College CP club
- r/leetcode, r/csMajors (Reddit)
- LinkedIn — CP/GSoC alumni se connect karo # Weekly Checklist Template

Har hafte is template ko copy karke apna progress track karo. Print karke bhi rakh sakte ho.

Week of: _______________

CP Practice

☐ Codeforces contest participate kiya (1/week minimum)
☐ ___ new problems solve kiye
☐ Upsolving kiya contest ke baad
☐ Team practice session (agar ICPC phase me ho): ___ hours

DSA/Topic Learning

☐ Is week ka topic: _______________
☐ Concept clear hai / Notes bana liye

Open Source / GSoC (jab applicable ho)

☐ Community me active raha
☐ Koi issue/PR pe kaam kiya
☐ Mentor se conversation hui

Interview Prep (Sem 4 ke baad)

☐ LeetCode: ___ problems solve kiye
☐ System design: ___ topic cover kiya
☐ Mock interview diya (haan/nahi)

General

☐ Rest days liye (burnout avoid karne ke liye): ___ din
☐ Progress kisi ko share kiya / accountability partner se baat hui

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Monthly Reflection (Mahine ke end me)

1. Is mahine ka biggest win kya tha?
2. Kaunsa area weak laga?

3. Agle mahine ka focus kya hoga?
4. Rating/metrics update:
    - Codeforces rating: ________
    - LeetCode total solved: ________
    - GitHub contributions: ________

Tip: Is file ko copy-paste karke har week ek naya entry banao, ya Notion/Google Sheets me convert kar lo tracking ke liye.